

MAINS MASTER PROGRAM (MMP) 2024

ENVIRONMENT-5

CONSERVATION AND EIA

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2. CONSERVATION: KEY GOVERNMENT INITIATIVES

1) NATIONAL ENVIRONMENTAL POLICY, 2006

- **Practice Questions:** Enumerate the key objectives of National Environment Policy, 2006. What are the key initiatives taken to achieve these objectives?
- **Introduction:** NEP, 2006 was formulated by MoEF&CC for the protection and conservation of environment. It replaced the 1986 policy.
- **The policy has 7 key objectives:**
 - **Conservation of Critical Environmental Resources**
 - **Intra Generational Equity - Livelihood Security for poor** - Poor communities, which are most dependent on environmental resources for their livelihoods, are assured secure access to these resources.
 - **Inter-Generational Equity** - To ensure judicious use of environmental resources to meet the needs and aspirations of the present and future generations.
 - **Integration of environmental concerns in Economic and Social Development**
 - **Efficiency in Environment Resource Use**
 - **Environmental Governance:** To apply principles of good governance (transparency, rationality, accountability, reduction in time and costs, participation, and regulatory independence)
 - **Enhancement of Resources for Environmental Conservation:** Resources include finance, technology, management skills, traditional knowledge, and social capital, for environmental conservation through mutually beneficial multi-stakeholder partnerships.
- **To achieve the above objectives, several initiatives have been taken** (see next topic for details)

2) KEY GOVERNMENT INITIATIVES FOR ENVIRONMENT CONSERVATION

- **Practice Questions:**
 - » Enumerate the key initiatives focused on environmental conservation in India [10 marks, 250 words]
- **Introduction:** The term 'Environment' was introduced in the Constitution of India through the 42nd Constitutional Amendment Act. This amendment added Article 48-A and Article 51-A(g).
 - » **Article 48-A** says that the state shall endeavor to **protect and improve the environment** and to **safeguard the forest and wildlife of the country**.
 - » **Article 51-A(g)** makes it a **fundamental duty** of every citizen of India **to protect and improve the natural environment** including forests, lakes, rivers and wildlife and to have compassion for living creatures.
- **Important Environment Conservation Efforts:**
 - » **National Environment Policy, 2006** formulated by MoEF&CC for the protection and conservation of environment identifies **7 Objectives**:
 - » **Laws which deal with Conservation of Environment**
 - Environment Protection Act, 1986;
 - Water (Prevention and Control of Pollution) Act, 1974;
 - Water Cess Act, 1977;
 - Air (Prevention and Control of Pollution) Act, 1981;

» **Laws in respect of forest and biodiversity are:**

- Indian Forest Act, 1927;
- Forest (Conservation) Act, 1980;
- Wild Life (Protection) Act, 1972: It provides framework for the protection of wild animals, birds, plants etc.
- Biodiversity Act, 2022

» **Project Tiger, Project Elephant etc.**

» **National Clean Air Program**

» **Namami Gange**

» **National Action Plan on Climate Change**

» **Swachh Bharat Mission**

» **National Green Tribunal:** Specialized judicial body to deal with environmental issues.

» **LiFE (Lifestyle for Environment):** It was unveiled by PM Modi at COP26. It aims to promote environment conscious lifestyle.

» **Panchamrita:** At COP26, India has given following targets to fight climate change:

- India will get non-fossil energy capacity to 500 GW by 2030.
- India will meet 50% of its energy requirements from renewable energy by 2030.
- India will reduce the total projected carbon emissions by one billion tonnes from now onwards till 2030
- India will reduce carbon intensity of its economy by less than 45% by 2030.
- India will achieve the target of net zero by the year 2070.

» **Digital Initiatives by MoEF&CC:**

1. **PARIVESH** (Pro-Active and Responsive facilitation by Interactive, Virtuous and Environmental Single Window Hub): It is a portal launched in 2018 which has simplified the process of environmental clearance. It has paved a way for constituting the centralized processing centre in the Ministry and has resulted in a paradigm shift in the environmental clearance process with hallmarks such as transparency, accountability, efficiency and consistency.
2. **Decision Support System (DSS):** It is a web-GIS application developed to provide qualitative and quantitative information with respect to forest area.
3. **Climate Change Knowledge Portal:** it is a single point resource which captures sector wise adaptation and mitigation actions that are taken by various ministries in one place. It helps in disseminating information among citizens.
4. **Wetlands of India Portal:** It is a single point source for information regarding wetland sites, projects, initiatives and trainings.

- **Conclusion:** Above are just few of the initiatives that India has launched for environmental conservation. Some of them like Project Tiger have been very successful and some have faced bureaucratic hurdles, resources constraints and poor implementation. Continuous efforts are needed to strengthen existing initiatives, initiate new innovative steps and foster greater public participation to achieve a cleaner and greener India.

3. ENVIRONMENTAL IMPACT ASSESSMENT

1) ENVIRONMENTAL IMPACT ASSESSMENT

- **PYQS** (see first booklet of Environment)
- **Other Practice Questions**
 - » Discuss the key challenges associated with Environmental Impact Assessment in India [10 marks, 150 words]
 - » "It is crucial to envision the EIA as anticipatory, participatory, and systematic in nature and a process that relies on multidisciplinary input, somethings that the draft EIA 2020 fails to do" Discuss [10 marks, 150 words]
- **Environmental Impact Assessment**
 - » EIA can be defined as the study to predict the **environmental, socio-economic, cultural and human-health impacts** of proposed project/activity. The global environmental law for the EIA is the "**precautionary principle**". Environmental harm is often irreparable so there should be a focus on prevention.
 - » It is a **decision-making tool** which **compares various alternatives** for a project and chooses the one which ensures best combination of economic and environmental costs and benefits.
- **Advantages of EIA:** By considering the environmental effects of the project and their mitigation early in the project planning cycle, environmental assessment has many benefits:
 - » Promotes environmentally safe and sustainable development.
 - » Optimum utilization of resources
 - » Saving of time and cost of the project
 - » Properly conducted EIA also **lessens conflict** by promoting community participation, informing decision makers, and helping lay the base for environmentally sound project.
- **History of EIA in India**
 - » The Indian experience of EIA started in **1976-77** when the Planning Commission asked the Department of Science and Technology to examine the river valley project from an environmental angle.
 - » **Till 1994**, EIA was an administrative decision and lacked statutory backing.
 - » **In 1994**, the Ministry of Environment and Forest, under the EPA, 1986, promulgated an **EIA notification making environmental clearance mandatory** for expansion or modernization of any activity or for setting up new projects listed in Schedule 1 of the notification.
- **EIA Notification, 2006**
 - » Notified by MoEF&CC under the Environmental (Protection) Act, 1986.
 - » The primary purpose of the notification is to assess the potential environmental impacts of developmental projects and ensure that they are carried out in environmentally sustainable and responsible manner.
 - » **Key Features:**
 - Unlike, the 1994 notification, it has put the **responsibility of clearing certain projects on the state government**. It **categorizes projects into two categories**:

- a. **Category A:** Require clearance at national level -> Projects are appraised by Impact Assessment Agency (IAA) and the Expert Appraisal Committee (EAC).
- b. **Category B:** Require clearance at state level. -> State Level Environmental Impact Assessment Authority (SEIAA) and State Level Expert Appraisal Committee (SEAC) provide clearance to category B projects.
 - **Category B1** (Mandatorily requires EIA)
 - **Category B2** (Doesn't require EIA)
- **Mandatory Clearance:** Projects falling in these two categories require prior environmental clearance from the respective regulatory authorities, before they can commence construction and operations.
- **Process of EIA:** It involves detailed study and assessment of the potential environmental and social impacts of the proposed projects. This include studying the baseline environmental conditions, identifying potential impacts, proposing mitigation measures, and preparing EIA Report.
- **Public Consultation:** It requires that all Category A projects undergo a public consultation process at both the draft EIA report stage and the final EIA report stage to allow affected communities and stakeholders to voice their concerns and suggestions.
- **Expert Appraisal Committee (EAC):** The MoEF&CC constitute EAC at the central and state levels to review the EIA reports and recommend or reject environmental clearance based on their findings.
 - » The EAC frames the scope of EIA study, and a preliminary report is prepared.
 - » The report is published, and a **public consultation** process takes place, where objections can be heard including from project-affected people.
 - » The EAC then makes a final appraisal of the project and forward it to MoEF&CC. The Ministry is ordinarily obliged to accept the decision of the EAC.
- **Post Clearance Monitoring:** To ensure that project complies with the conditions specified in the environmental clearance and that any adverse environmental impacts are mitigated.
- **Violence and Penalties:** Violations may lead to penalties and even legal action against the project proponent or developers.
- **Some general shortcomings of EIA in India**
 - i. **Delays:** EIA has remained a time consuming process thus leading to delay in the project cost.
 - ii. **Lacks Exhaustive Coverage:** EIA is not mandatory for several projects which are small investment or are not listed in Schedule 1. these projects also have severe environmental impact.
 - iii. **Public hearing** comes into play at later stages, rather than at the beginning. Public hearing at the beginning could reduce a number of conflicts at later stage. Further, for several types of projects public hearing is not mandatory.
 - iv. **Quality of EIA** is also questionable. On many occasion, the EIA report has false or incomplete data. **Consultant agencies** are rarely held accountable.
 - v. **Credibility crunch:** Since the preparation of EIA for the project is the responsibility of the project promoter, there are several conflict of interests involved.
 - Since EIA consultants are paid by the project proponent, there is less chance that they will go against them.
 - The team formed for conducting EIA studies is lacking the expertise in various fields such as environmentalists, wildlife experts, Anthropologists and Social Scientists.

- **Further, accreditation of EIA consultants is missing.**
- vi. **Poor Compliance Monitoring: Lack of administrative capacity to ensure compliance** often renders lists of clearance conditions meaningless.
- vii. **Increased tendency to exempt projects from prior clearance:** For e.g. as per the EIA rules notified in 2022 have exempted different projects from prior clearance.
- **Way forward: Improvements**
 - » **Increase the Coverage of EIA:** Every project which has substantial impact on environment should be brought under the EIA regime.
 - » **EIA Clearance should be conditional:** There should be a provision of automatic withdrawal of clearance if the condition of clearance are violated.
 - » **Set up an Independent EIA Authority** - to increase credibility of the EIA process.
 - » **EIA consultants** should be paid by MoEF&CC or some pool on which the project proponent shouldn't have control.
 - » Increase the **scope of public hearing** -> Make it happen in the beginning and increase its coverage for more types of projects.
 - Local communities should have all the information - from notification to project clearance - available with them to take better decisions.
 - » **Diversify the National Green Tribunal** by inclusion of more members from the field of environment.

A) AMENDMENT TO EIA RULES NOTIFIED (JULY 2022)

- It has exempted highway projects of strategic and defence importance, which are 100 km from the LoC, among other locations, from an environmental clearance before construction.
- Thermal power plants upto 15 MW based on biomass or non-hazardous municipal waste using auxiliary fuel such as coal, lignite or petroleum products up to 15% have also been exempted - as long as the fuel mix is eco-friendly.
- Increasing the threshold of ports which exclusively deal in fish handling and caters to small fisherman, which are exempted from environment clearance.
 - » This is considering issues of livelihood security of fishermen involved at fish handling ports and harbors, and less pollution potential of these ports and harbors.
- **Toll plazas** that need more width for installation of toll collection booths to cater to a large number of vehicles, and expansion activities in existing airport related to terminal building expansion without increase in airports existing area, rather than expansion of runways, etc., are two other exempted projects.

B) THE SC HAS SET ASIDE A NOTIFICATION ISSUED BY THE MOEF&CC IN 2020 THAT EXEMPTED EXTRACTION OF ORDINARY EARTH FOR LINEAR PROJECTS, SUCH AS ROADS AND RAILWAY CONSTRUCTION, FROM OBTAINING ENVIRONMENTAL CLEARANCE (EC) (MARCH 2024)

- **Background:**
 - » In Sep 2006, MoEF&CC notified activities that would require prior EC.

- » In Jan 2016, another notification was issued which exempted certain category of projects from this requirement.
- » In March 2020, third notification added "Extraction or sourcing or borrowing of ordinary earth for the linear projects such as roads, pipelines etc." to the list of exempted activities.
- **Why the exemption?**
 - » The general purpose of the exemption was to conform to the amendments made to the **Mines and Minerals (Development and Regulation) Act, 1957** in March 2020, allowing new lessees to continue mining for two years with the statutory clearance and license issued to their predecessors.
 - » The Centre also argued before the NGT that the exemption was necessary "for the aid of general public", and would help "the Kumhars (potters), farmers, gram panchayats, vajaras, oads of Gujarat".
 - » Government also said that grant of exemption was a **policy matter** that didn't warrant judicial interference.
- **Challenge to the Exemption:**
 - » The exemption was challenged before the NGT on the ground that allowing the extraction of earth indiscriminately was arbitrary and violative of Article 14 of the Constitution of India.
 - » The petitioner also argued that the exemption violated the Supreme Court verdict in **Deepak Kumar versus the State of Haryana (2012)** which required for prior EC in the leases.
 - » Ministry had also "circumvented the legal procedure of inviting public objections before issuing the 2020 notification by wrongly exercising its powers to do away with such requirements "under the garb of 'public interest' during the COVID-19 national lockdown.
- **NGT in Oct 2020**, asked government to "revisit" the notification within three months and held that the ministry should strike a balance and instead of being blanket exemption, it needs to be hedged by appropriate safeguards.
- **Center sat on NGT order** until the matter went to SC. After SC concluded the hearing and reserved the judgement, the MoEF&CC notified that exemption in question would be **subject to the compliance of SOP and environmental safeguards** issued in this regard from time to time.
- **Supreme Court Verdict:**
 - "**Completely unguided and blanket exemption**" was arbitrary and violative of Article 14 because the 2020 notification didn't even define linear projects, or specify the quantum and extraction area. This defeats the purpose of EP Act.
 - The court held that the Ministry offered no justification for concluding "that in the public interest, the requirement of public notice should be dispensed with" at any stage - neither in the notification itself or in its submissions to the NGT and SC.
 - Even the Aug 2023 notification failed to elaborate on the concept of linear projects, specify the authority responsible for environmental safeguards, restrictions on the quantum of extraction etc.
 - The court also said that it failed to understand the **undue haste shown by Central government in issuing the impugned notification** which was issued two days after the nationwide lockdowns was imposed.

- **Past cases where these kinds of exemptions have been brought under Judicial Scrutiny:**
 - In July 2021, through a notification Ministry sought to perpetuate an amnesty window opened for just six months in March 2017 to clear projects under the "violation category" and issued ex-post facto approval to more than 100 projects, until **the Supreme court stayed it in Jan 2024**.
 - In March 2024, the High Court of Kerala quashed a 2014 notification that exempted institutions and industrial sheds with build-up areas of more than 20,000 sq m from obtaining EC.

2) GREAT NICOBAR ISLAND PROJECT

- **Why in news?**
 - » NGT constitutes high powered committee to revisit environmental nod granted to Great Nicobar Island Project (April 2023)
- **Island of Great Nicobar:**
 - » The Island of Great Nicobar is the southernmost tip of India, and a part of the A&N archipelago. It is hilly and covered with lush rainforests that are sustained by around 3,500 mm of annual rainfall. They sustained numerous endangered and endemic species including the giant leatherback turtle, the Nicobar megapode, the Great Nicobar Crane, the Nicobar crab eating Macaque, and the Nicobar tree shrew. It is also home to **two tribal communities** - the Shompen and the Nicobarese.
- The Great Nicobar Island Project (GNI) is a mega project to be implemented at the southern end of the Andaman and Nicobar Islands. The project is estimated to cost Rs 72,000 crores and has four components:
 - » A Rs 35,000 crore transshipment port at Galathea Bay
 - » A dual-use military-civil international airport
 - » Township development
 - » 450 MW Gas and Solar based power plant
 - » The project is to be undertaken by a government undertaking called the **Andaman and Nicobar Islands Integrated Development Corporation (ANIIDCO)**.
- **Positives of the Project:**
 - » **Holistic development** of A&N Islands.
 - Allow Great Nicobar to participate in the regional and global maritime economy by becoming major player in cargo transshipment.
 - Better Infrastructure (port, airport, electricity) will also attract tourism.
 - » **Ease of living:** Better infrastructure and connectivity will also simplify lives of communities settled on the A&N Islands.
 - » **Strategic significance:**
 - A&N is already the place of India's first joint/unified operational command - the A&N command in Port Blair.
 - Having a military airport on Great Nicobar, so close to Malacca strait, will give India military advantage in case of conflict with China.

- Only 31 of archipelago's islands and islets are inhabited. This means there is a possibility of surreptitious occupation - a la "Kargil Heights" - by a covetous neighbor.
 - It also provides an opportunities for cooperation and collaboration with other like minded countries in the region.
- » Ministry of Tribal Affairs have also said that the project will not lead to displacement of tribal people.
- **Criticism:**
 - » **Negative Impact on Biodiversity:** The region is a **biodiversity hotspot** with tropical evergreen trees. The region has many rare and unique species and some even undocumented.
 - For e.g. The Galathea bay in the island is a nesting ground for birds. It is also the one of the world's largest nesting ground for the **giant leatherback sea turtle** (which is under Schedule-1 of WPA)
 - In fact, in Jan 2021, the Indian government denotified two wildlife sanctuaries - the Galathea Bay Wildlife Sanctuary and the Megapode WLS.
 - » **Deforestation:** The project will require diversion of 130 sq km of forest land and the felling of around 10 lakh trees.
 - » **Violates various laws and rules:** For e.g. project area is part of CRZ 1A and 1B, where developmental activities are highly restricted.
 - » **Threat to tribes:**
 - **Health:** For e.g. Shompen tribe has very little contact with the outside world and thus may not have immunity to infectious diseases.
 - **Displacement:** Some Shompen settlements also overlap with the areas the NITI Aayog has proposed to be used for the transshipment terminal.
 - » **Disaster Prone region:** The A&N Islands archipelago is located in the "ring of fire" thus vulnerable to high intensity earthquake. It is in seismic Zone-V.
- **Way Forward:**
 - **A comprehensive and transparent environmental impact assessment** should be done to ensure that negative consequences are reduced to minimum.
 - **Identify the least harmful methods** and work with them, even if the cost of the project increases a bit
 - **Keep very high diversity areas untouched.**
 - **Consult local communities and stakeholders** to address their concerns and garner their support for the project
 - **Strengthen mitigation measures** like afforestation, coral reef transplantation, and creation of biodiversity park.
- **Conclusion:** The Great Nicobar Project is a complex endeavour with huge benefits and potential environmental risks. It's important to do a thorough assessment to ensure that benefits are realized in long run without negatively impacting the ecology of the region.

3) HIMALAYAN ECOSYSTEM AND KEY THREATS

- **Example Question:**

- » An explosive cocktail of climate change and aggressive road and dam building in the geologically unstable Himalayan ranges bring its own environmental, social and economic challenges. Discuss [10 marks, 150 words]
- » Discuss the key factors behind the rise of Tourism in Himalayas in recent years. How has increase in tourism impacted Himalayan ecosystem [15 marks, 250 words]
- **Introduction-1:** The Indian Himalayan region is spread across 13 states/Uts. Around 50 million people live in this region and have diverse demographic, and versatile economic, social and political systems.
- **Introduction-2:** The Teesta dam breach in Sikkim, devastating floods and landslides in Himachal are a stark reminder of the havoc development model is wreaking on our environmental and ecology, specially in the mountains.
- **Significance of Himalayas**
 - » **Environmental:**
 - **Biodiversity:** Himalayas are one of the biodiversity hotspots in the country and thus is crucial for biodiversity protection.
 - **Forest cover:** More than 40% of Himalayan geographical area is under forest cover. This is 1/3rd of the countries total forest cover. These forests are a major carbon sink in the country and also avert soil erosion here.
 - **Water reservoirs of the country:** Annually, 1,200 billion cubic meter of waters flow through the Himalayan rivers. More than 50% of India's population depend on these rivers.
 - » **Economic:**
 - Himalayas are an important source of fruits like Apples, Oranges, etc which also has a lot of export valuation. Further, Himalayas are also the most important timber producing region in the country.
 - **Tourism:** Himalayas provide a major destination for tourism in the country and brings a major chunk of revenue for Himalayan states.
 - » **Cultural & Religious:** Himalaya is considered the home of deity Shiva and for centuries Hindus from across the country have visited many shrines, temples and sanctuaries located on the peaks. Himalaya has also acted as a important barrier and have led to development of unique cultures on north and south of it.
 - » **Geographical:** Himalayas, protect Indian sub-continent from cold winds from further north.
- **Key factors impacting Himalayan Mountain Ecosystem**
 - **Unscientific Developmental activities:** Dams, Roads, Bridges etc. without proper EIA
 - **Large Road projects (like the Char Dham Highway),** which involve widening of roads leads to cutting of trees and weakening of mountain ranges. This leads to increased vulnerability to floods, landslides etc.
 - These activities, along with deforestation, mining etc. are making Himalayas more vulnerable to landslides and floods.
 - **Invasive species** like Lantana Camara, Parathenium hysterophorus etc have been reported in most of the Indian Himalayan region.
 - **Climate Change** is roasting the Himalaya region, threatening millions
 - Temperature in the Himalayas have risen faster than in the other mountain ranges

- Scientists say that Himalayan glaciers will recede by a third by 2100 even if the increase in global temperature is capped at 1.5 degree C.
- Forest fires, floods etc.
- **Increase in Tourism, Trekking etc -> Pollution and Growing Garbage Pile:**
 - More and more Indians are going as tourists to Himalayas, but the trash that they are leaving behind on the mountains are spelling disaster for environment.
 - **Trekking still remains an unorganized sector**
 - **Implementation of government norms** like 'No trace' policy is weak.
 - **Why increase in Tourism to Himalayas:** Rising disposable income; Increasing younger population; Availability of good equipment, guides etc; Rising awareness about trekking and other Himalayan adventure through social media. Etc.
 - **Way Forward:**
 - **Penalties** - On erring trekkers based on 'Polluters' pay principle
 - **Awareness and sensitizing Program** for both trekkers, operators and hill communities
 - **Waste Management Infrastructure** in the Himalayan region needs to be strengthened to deal with the garbage left behind.
- **Geopolitics** -> Tensions between India and China, India and Pakistan etc. is intensifying fast track developmental projects in the region, further increasing the vulnerability to disaster.
- **Way Forward**
 - **Monetize Ecosystem services provided by these states: 'Green Bonus'** to Himalayan states from Union Government needs to increase, so that these states don't promote unsustainable economic activities and tourism to mobilize resources.
 - **Improved, more stringent EIA norms** for Himalayan region as it is more fragile and vulnerable and thus requires bigger protection.
 - **Improving EIA system** in the country to control the harmful impacts of large projects like Dams, Road projects etc. For Himalayan region a **more stringent EIA mechanism** should also be considered.
 - **Afforestation and Mitigating Climate Change** -> work towards achieving the Paris Agreement targets
 - **Sustainable Tourism** - Control the number of people visiting various tourist sites, sensitize them about sustainable tourism and increase penalty on violators.
 - **Increased International Cooperation** - Reduction in political tension and increased cooperation among Himalayan nations is crucial to ensure joint efforts towards protecting Himalayan ecosystem.

4) MINING – ENVIRONMENTAL IMPACT

- **Introduction:** The death of 10 people in July 2024, while working in illegal coal mines in Gujarat, have again highlighted the safety, environmental and economic concerns associated with mining sector in India.
- **Key issues created by mining sector:**
 - i. **Deforestation and Loss of Biodiversity:** Large scale mining operations often require the clearing of forests and other natural habitats, leading to loss of biodiversity and disruption of ecological balance.
 - For e.g. In Hasdeo mines in Chhattisgarh, over 2 lakh trees have been marked for felling in 2022
 - ii. **Soil Erosion, Degradation and Pollution:** Removing vegetation and disrupting soil during mining can result in **increased soil erosion** and degradation. It can lead to loss of fertile topsoil and can disrupt natural nutrient cycle. Further, mining may also add heavy metal to the soil causing soil pollution.
 - iii. **Water Pollution** (including ground water pollution):
 - **Acidification of nearby water bodies** due to washing of sulfur rich compounds in the rock after it is broken and stored near the mines. This acid drainage has negative impact on environment and biodiversity of the region.
 - **Old and abandoned mines** have harmed water quality long after mining has ceased.
 - iv. **Air Pollution**
 - Opencast coal mining creates air pollution problem in respect of dust and the fines which contain coal particles, benzene soluble matters etc.
 - This particulate matter pollution is a threat to human health and environment.
 - v. **Noise Pollution** - For e.g. coal mining operations are generally associated with high noise pollution.
 - vi. **Altering of geology, topography, hydrology and ecology of sites**
 - Mining have also left thousands of square kms of land altered. In some cases, particularly mountaintop removal mining, entire land forms are permanently shaped.
 - **Erosion and Sedimentation** have a devastating effect on environment of the place where coal mine is located.
 - Illegal sand mining pose a threat to the **riverbeds, natural flow of water and fishes**.
 - vii. **Threatens Biodiversity**
 - A research by NGT committee found that illegal sand mining threatened to further degrade riverine habitat that had already deteriorated. If nesting and basking sites are lost, rivers will no longer be a sanctuary for riverine species.
 - For e.g. Illegal Sand mining on Son River has led to drastic reduction in the population of Gharials over the years.
 - viii. **Economic Losses:** Illegal mining also leads to loss of royalty revenue.
 - ix. **Social Impact:** Mining activities also lead to displacement of local communities and disrupt traditional livelihood. It may also cause social issues such as conflict over land rights etc.
- **Recent Steps to Improve the Situation:**

- » Ministry of Coal sets up sustainable development cell for environment mitigation measures:
 - It is focused on promoting environmentally sustainable coal mining in the country; it addresses the environmental concerns during the decommissioning or closure of mines
- » **Steps by Coal India to reduce air pollution in transportation:**
 - **Low capacity tippers** (10-12 tonnes) are being replaced by **large tippers (35 tonnes)** to reduce the number of trips leading to reduction in the air pollution.
 - **Coal carrying trucks** are optimally loaded and covered with tarpaulin.
 - **Mobile Water sprinklers** are deployed along the haul roads.
 - **Deployment of road sweeping machines** to control dust pollution
 - **Tree Plantation** is being promoted to minimize soil erosion; air pollution and noise pollution.
 - In last five years (FY 2016-17 to FY 2020-21), CIL has planted more than 94 lakh of saplings over an area of about 3,873 Ha in inside mine lease area and around 7.13 lakh sapling over an area of about 500 Ha.
- » **Checking illegal mining:**
 - Concrete walls have been erected on the mouth of the abandoned mines to prevent access and illegal activities.
 - **Surprise raids/ checks; check posts at vulnerable points;**
 - **Training** of security/CISF personnel,
 - **Dumping of the overburden** is being done on the outcrop zones
 - **Increased cooperation** with state governments
- » **Better Monitoring and Surveillance: Mobile App - "Khanan Prahari"** and one **web app Coal Mine Surveillance and Management System (CMSMS)** for reporting unauthorized coal mining activities so that monitoring and taking suitable action on it can be done by Law & Order authority.
- » **Coal India Limited (CIL)** takes Out of Box Initiative to Produce **Sand from Overburden** at a much cheaper Price
- **Way Forward**
 - » Strengthen the provisions of Environmental Impact Assessment (covered separately in details)
 - » Ensuring effective oversight and long-term monitoring.
 - » Hold companies accountable for damages
 - » Reduce, Recycle and Reuse
 - » Making mining process more efficient -> International Collaboration; Technological advancements etc.

5) SAND MINING FROM RIVERS

- **Introduction:**
 - » The sand and gravel are one of the most important construction material. With changes in socio-economic circumstances in the country, the demand for both sand and gravel is increasing. This has led to unsustainable extraction of sand from the river banks .
- **Negative Impact of Sandmining:**
 - » **Negative impact on biodiversity:**
 - The eggs, young and seed as well as young plants are trampled under the wheels of the vehicles that collect sand.

- Studies in Chambal shows that sand mining caused human intrusion into basking sites, nesting patterns, and egg laying of gharials and freshwater turtles, causing them to leave the area.
- » **River Bank Erosion**
- » **Subsidence:** Sand mining can lead to land subsidence, which is the sinking of the ground surface.
- » **Decline in ground water:** Reduced water retention capacity; Disruption of natural hydrological cycle; surface water runoff (without sand being there to slow down the flow of water)
- » **Salt Water Intrusion:** In coastal regions, excessive sand mining may lead to salt water from ocean entering into rivers and other water bodies.
- **Way Forward:**
 - » **Strengthen Environmental Regulations and Enforcement:** The current regulations on sand mining in India are weak and poorly enforced. This has led to widespread environmental damages.
 - Effective monitoring -> increasing number of government officials.
 - Impose stricter penalties.
 - » **Sustainable Mining Practices:** This could involve using less water, minimizing the impact on riverbeds etc.
 - » **Promote Alternate Materials** - there are a number of alternative materials that can be used in place of sand, such as recycled concrete, manufactured sand (M-Sand), and geopolymers.
 - » **Improve Construction Practices:** For e.g. builders can use recycled concrete and other alternative materials, and they can design buildings which require less sand.
 - » **Education and Awareness:** This can help in promoting policies that protect environment from sand mining.
 - Make it easier for citizens to report illegal sand mining activities.
 - » **Reforming and strengthening Environmental Impact Assessment** system.

6) HYDROPOWER PLANTS – ENVIRONMENTAL IMPACT

- **Introduction:** Death and devastation caused by the bursting of Teesta Dam project in Oct 2023, has again highlighted that while hydropower projects are important for development for the country, if not properly built and managed, they may become devastating.
- **Significance of Hydropower plants**
 - » **Clean Energy** -> Electricity generation.
 - » flood control.
 - » irrigation.
 - » Water Security (specially for a Monsoon type of climate region)
 - » Geo-political implications
- **Concerns raised:**
 - » **Himalayas are Earthquake prone zones** -> Dams may worsen the earthquake disaster with floods (e.g. the 2015 Nepal Earthquake damaged several dams)
 - » **Dams aggravate other disasters:**
 - **Ravi Chopra Committee** formed after the Kedarnath floods in 2013 said that dams had directly and indirectly aggravated the 2013 Kedarnath disaster.
 - **Amelie Huber**, an ecologist who has done extensive research on the hydropower development in north-east India, say that dams in mountain regions that are exposed to earthquakes, floods, extreme rainfall, avalanches and landslides, are "risk laden artifacts"
 - » **Large scale projects** weaken the mountain system and increase vulnerability to landslides, avalanches etc. to which Himalayas are already highly vulnerable.
 - » **Silt deposition** -> Reduces the water carrying capacity of dam in long run; reduces the fertile silt going to lower levels.
 - » **Hampers Ecological Flow** in rivers which hampers river's ecological services.
 - » Further, global warming and glacial melts are increasing the vulnerability to GLOF which would further make these projects vulnerable.
 - » These large hydroelectric projects also cause loss of forest cover and biodiversity in Himalayas, which are one of the global biodiversity hotspots.
 - Several international studies have shown that dams are responsible for loss of more than 80% of freshwater species since 1970.
 - » **Need of relocating people** further complicates things and brings its own social and economic challenges.
- **Way Forward:**
 - » **Strengthen Environmental Impact Assessment norms** (discussed separately)
 - Adopt and implement policies prohibiting the construction of dams in protected areas or free-flowing rivers or in higher altitude of mountains.
 - Avoid projects with irreversible impacts on critically endangered species
 - Requiring the free, prior and informed consent of indigenous people on projects that will impact them.
 - » **Focus on Micro and Small-Hydro**
 - Microhydel projects would have less adverse social and environmental impact and are suitable and successful technological and developmental interventions.
- **Conclusion:** There is a requirement of bringing a balance between environmental requirements and developmental needs to ensure long term sustainable development.

4. CORALS

- **Why in news?**
 - Close to 75% World Coral Reefs are exposed to bleaching - level heat stress in last 17 months (July 2024: Source DTE)
 - Catastrophic bleaching event: 97% of coral at Australian island wiped out in 3 months (June 2024)
- **Example Questions:**
 - Why is it important to protect global Coral Reefs? What are the key challenges faced by Corals across the world? [15 marks, 250 words]
- **What are Corals and Coral Reefs?**
 - » Corals are colonial organisms made up of Individual polyps. Coral 'Polyps' are tiny animals related to **anemones and jellyfish**.
 - » Coral Reefs are larger underwater structure composed of the skeletons of Corals. Reefs are built by coral polyps as they secrete layers of Calcium carbonate from under their skin. These skeletons made from calcium carbonate, protect the coral animals from predators and also offer a substrate on which new Coral Polyyps can attach themselves.
 - » **Coral** reefs grow best in warm water and they prefer a shallow range with lots of sunlight for their symbiotic algae.
- **Significance of Coral Reefs**
 - » **Biodiversity Protection:** Coral reefs are believed to have highest biodiversity of any ecosystem on the planet - even more than the tropical rain forests. They occupy less than 1% of the ocean floor but is home to 25% of marine life. They are sometimes also known as '**The tropical rainforest of the Oceans**'. They also provide substrate for mangroves.
 - » **Climate Regulation:** Coral reefs play an important role in absorbing carbon and play an important role in fighting global warming.
 - » **Disaster Mitigation:** Corals provide protection from storms and Tsunamis by breaking the large waves.
 - » **Economic Benefits:** The value of goods and services provided by Coral reefs is estimated to be \$2.7 trillion per year. They also provide revenue from fishing and tourism.
 - » They are also the largest biogenic CaCO₃ producer.
 - » **Other Advantages:**
 - **Medical Potential:** Coral reef organisms are being researched for the potential of developing treatment for cancer, arthritis etc.
- **Threats:** As per Global Coral Reef Monitoring Network (CCRMN), in the last decade, world has lost 14% of its Coral.
- **Key Factors:**
 - i. **Man-Made Causes**
 1. **Pollution, Ocean Acidification, etc.**
 - Eutrophication -> Deoxygenation -> Dead zones. As per a recent study, deoxygenation has emerged as the biggest threats in recent years.
 2. **Overfishing, Unsustainable Tourism, and Poor Coastal Management**

3. Mechanical Damages

4. Thermal Pollution

5. Climate Change

- As per the GCRMN report, the reefs all over the world are under relentless stress due to warming caused by climate change.
- Higher sea surface temperatures have also become a factor for coral bleaching.

ii. **Natural disturbance** such as violent storms, El Nino Southern Oscillation, epizootics etc.

- Coral Bleaching

- » When **Corals are stressed by changes in conditions** such as temperature, light, or nutrients, they **expel the symbiotic algae** living on their tissues, causing them to turn **completely white**.
- » It has emerged as one of the major reasons for coral reef destruction. For e.g. in 1998, it caused the loss of 8% of the world's corals.



- Way Forward:

- » **Sustainable Development**
 - Fighting Pollution
 - Mitigating Climate Change
 - Restricted mining, tourism and fishing in Coral areas
- » Creation of more Marine Protected areas (High Seas Treaty)
- » Increased international cooperation.

5. MANGROVES

- **Example Questions:**
 - » Discuss the significance of mangroves for India's coastal Ecosystem. How are mangroves affected by Climate Change? [10 marks, 150 words]
- **Introduction**
 - » Mangroves are a group of about 70 species of trees, shrubs, and ferns. It shows convergent adaptations to saline, oxygen deficient, habitats. The 'true mangroves' exhibit all or few of the typical mangrove adaptations:
 - **Tolerance to Salinity; Salt filtering or exuding system**
 - **Tolerance to oxygen deficient soil**
 - **Stilt or knee roots, Aerial Roots** (pneumatophores)
 - **Note:** Aerial roots project above the mud and have small openings (lenticles) through which air enters, passing through the soft spongy tissues to the roots beneath the mud. This acts as site of oxygen intake for the submerged roots.
 - **Succulents (thick fleshy leaves)**
 - **Leathery leaves**
 - **Viviparous Seedlings**
 - Vivipary is a phenomenon that involves seeds germinating prematurely while they are still inside or attached to the parent plant or fruits. Many mangrove species show vivipary. The ovum is fertilized while still on the parent tree and grows by a combination of photosynthesis and acquisition of nutrients from the parent until it may reach a length of 50 cm.
 - These features allow these species, belonging to different families and genus survive along the coasts. Some prominent mangrove genera are Avicennia, Rhizophora, Sonneratia, Brugueira, etc.
- **Importance:**
 - **Protection against Tsunamis, Storm Surges and Soil Erosion**
 - **Enhance Sediment deposition:** Therefore, they act as a zone of land accretion due to trapping of fine sediments.
 - **Reduces sea water pollution**
 - **Biodiversity:** They act as fertile breeding ground for many fish species and other marine fauna.
 - **Economy:** They act as important source of livelihood for the coastal communities dependent on collection of honey, tannins, wax and fishing
 - **Fight Climate Change:** Important carbon sink
- **Factors harming Mangrove Ecosystem:**
 - **Land Reclamation for agriculture**
 - **Industrialization along the coastlines**
 - **Discharge of untreated domestic sewage and industrial effluents**
- **Threats:**
 - **Natural Factors:**

- » Climate Change -> more intensive cyclones
 - Warming ocean waters also reduce the dispersion of the seeds as warm waters are less buoyant and thus seeds sink to the ground rather than traveling long distances.
- » Infestation by barnacles
- » Insect Pests such as wood borers, caterpillars
- **Anthropogenic Threats:**
 - » Unplanned Urban Development and industrialization
 - » Wood and wood products; Grazing by castles.
 - » Reclamation for agriculture and aquaculture
 - » Pollution
 - » Poor implementation of CRZ norms
 - » **Invasive Species**
- **Way Forward:**
 - » Identify the vulnerable zones (mapping)
 - » Improved Monitoring -> land and aerial survey
 - » Strict implementation of Coastal Regulation Zone norms
 - » Protection -> declaration of WLS, NP etc.

A) MISHTI (MANGROVE INITIATIVE FOR SHORELINE HABITATS & TANGIBLE INCOMES)

- » It is a centrally sponsored scheme launched by MoEF&CC in 2023-24.
- » It aims to:
 - **Increase the mangrove cover** in India by 540 sq km along the coastline and on saltpan lands in 9 coastal states and 4 Union territories during five years commencing FY 2023-24 onwards.
 - **Conserve and restore mangrove ecosystem**
 - **Promote ecotourism and Livelihood generation** activities in mangrove areas.
- » **Components:**
 - Mangrove Plantation
 - Awareness Generation
 - Capacity Building
 - Research and Development

B) MANGROVE ALLIANCE FOR CLIMATE

- » **Details**
 - Mangrove Alliance for Climate (MAC) is an initiative led by UAE and Indonesia. It also includes India, Sri Lanka, Australia, and Spain.
 - It seeks to educate and spread awareness worldwide on the role of mangroves in curbing global warming and its potential as a solution for climate change.
 - The intergovernmental alliance works on a voluntary basis meaning that there are no real checks and balances to hold the members accountable. Instead, the parties will design their own commitments and deadlines regarding planting and restoring mangroves.
- » **Suggestions: Integration of Mangroves** into the national REDD+ programs need of the hour

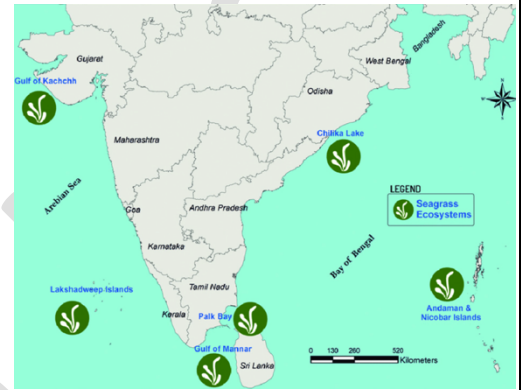
6. SEA GRASS

- Practice Questions

- » Discuss the key ecosystem services provided by seagrass. Explain the factors responsible for the degradation of seagrass habitats globally. Suggest key measures to reverse this degradation [15 marks, 250 words]

- Introduction:

- » Seagrasses are marine flowering plants that thrive fully submerged in shallow oceanic and estuarine habitats. They are one of the most important coastal habitat.
- » **Global** coverage of seagrass is estimated to be $3.45 \times 10^5 \text{ km}^2$, which represent **about 0.1% - 0.2%** of the ocean floor.
- » In India, total seagrass cover is estimated to be 517 km² with 14 reported species and six genera.
- » **Distribution of seagrasses in India:** Gulf of Mannar & Palk Bay (80 and 320 KM²); Chilika Lagoon (20 KM² to 80 km²); Gulf of Kutch; Lagoons of Lakshadweep; A&N Islands



- Significance of Seagrasses Ecosystem:

▪ Provisioning Services:

1. **Medicine** (treatment of heart conditions, sea sickness etc.)
2. **Food** (Nutritious seeds)
 - Recently, a study has shown the presence of various biological metabolites in some India seagrass that can be used effectively in the food and pharmacological industries.
3. **Fertilizer** (Nutrient rich biomass)
4. **Livestock feed** (food for goats sheep etc.)
5. **Building Material** (Such as roofing for houses)

▪ Supporting Services:

1. **Key Fishing Grounds** as they offer a complex habitat for a variety of fish and other marine organisms like Dugongs.
 - Seagrass based fisheries are globally important and are present wherever seagrass exists, supporting subsistence, commercial and recreational activities.
 - Their high rates of primary production result in well-oxygenated waters that support complex food webs.

▪ Regulating Services:

1. **Coastal Protection:** Seagrass reduce the energy of waves and thus protect the seashore.
2. **Carbon Sequestration:** Seagrass store more than twice as much carbon from planet warming CO₂ per square mile than forests do on land, according to a 2012 study. They accumulate CO₂ from both in-situ production and sedimentation of particulate carbon from the water column.

3. **Water Purification:** They trap fine sediments and suspended particles in the water column and increase water clarity.

- **Threats to Seagrass Ecosystems:** Globally, seagrass habitations have declined in area and several species are threatened due to several natural and anthropogenic stressors:

- **Natural Stressor:** Cyclones, heavy rainfall, coastal uplift and subsidence, grazing herbivores, and diseases
- **Anthropogenic Stressors:**
 - i. **Commercial Fishing and trawling activities:** this is the most important threat to seagrass in India.
 - ii. **Boat activities** for recreational purposes
 - iii. **Runoff (Pollution)** from coastal aquaculture and agriculture
 - iv. **Shell Harvesting and Seaweed cultivation**
 - v. **Accidents** like Oil Spills.

- **Key Steps** which protect Seagrasses:

- » **CRZ Notification** 2011, issued under Environmental (Protection) Act, 1986, has classified seagrass meadows as CRZ1-A (Ecologically sensitive area). It prohibits developmental activities in its vicinity.

- **Way Forward:**

- » **Mapping of existing seagrass areas** and areas with potential of growth of seagrass.
 - Here steps can be taken for natural expansion of seagrass.
- » **Effective implementation of available legislation** may help reduce the threats.
 - Ensuring that the provisions of protection accorded to CRZ-1 are strictly followed can prevent degradation and can lead to self-restoration of seagrass meadows.
- » **More study** on the various uses of seagrass by local communities and then spreading awareness among all the stakeholders about the various ecosystem services provided by the seagrasses, especially their role in carbon sequestration.
- » **More R&D** on the various applications of seagrass by local communities
- » **Research** is also needed on the techniques for seagrass transplantation to increase the coverage of seagrass.
 - Learn from **Seastore Seagrass Restoration Project** on how to train local population to autonomously transplant and increase seagrass cover.
- » **Increased collaboration between various stakeholders** - Government, researchers, local communities etc to develop an action plan for seagrass conservation.

7. NOISE POLLUTION

- **Practice Question:**
 - » What is Noise Pollution? Discuss the most important sources of noise pollution in Indian cities [10 marks, 150 words]
 - » Critically examine the legal and regulatory framework to control noise pollution in India. Suggest key reforms to remedy the problem of noise pollution [15 marks, 250 words]
- **Intro**
 - » Noise pollution refers the presence of such levels of noise or sound in the environment that are disturbing, irritating and annoying to living beings.
- **Causes of Noise Pollution:** Vehicles, factories, industries, construction sites, firecrackers, loudspeakers, domestic appliances; TV/Radio etc.
- **Effects of Noise Pollution**
 - » **Negative Health Impact:**
 - **Physiological damage:**
 - Sleeplessness, headache, difficulty in talking;
 - Loss of hearing and deafness
 - Cardiac Disturbance
 - **Increase changes Diabetes and Hypertension**
 - **Psychological damage:**
 - Stress, tension, aggressiveness.
 - Mental Imbalance and nervous debility
 - **Children, elderly and Marginalized community** are more vulnerable
 - » **Negative Impact on Biodiversity:**
 - It may disrupt animal communication, breeding patterns, migration routes of animals.
 - It may also cause **growth and development issues**. For e.g. a recent study published in Conservation Biology Journal noted that chicks of the birds which were exposed to noise were smaller than the ones in quiet nests.
 - » **Negative Economic Impact:**
 - **Value of property** in high noise area reduce.
 - **Healthcare cost** goes up
- **What is the safe limit for noise?**
 - » As per the 2018 WHO guidelines a health protective recommendation for road traffic is **53 dB**.
 - » Hearing loss begins if a person is exposed more than 8 hours a day to a noise level of 80-90 dbA.
 - » A level of 140 dbA is painful and 180 dbA could even kill a person.
 - » **What is the level of noise pollution in India? -> Steadily increasing.**
 - Advanced sound systems used in public meetings, festivals, marriage reception, concerts etc.
 - **Automobile horns and traffic** movement
- **What is being done to curb the noise pollution?**
 - » **Institutional Mechanism:** **CPCB** is mandated to track noise levels, set standards and via state units ensure that excessive noise is controlled.

- » **Supreme Court Verdicts:** Since 1980s and 90s, there are several court judgments restricting the noise pollution from industries, firecrackers etc.
 - **2015 Supreme Court Judgements:** In 2015, the Supreme Court, acting on a petition filed by four infants (all aged between six months and 14 months) seeking curbs on air and sound pollution, banned the bursting of sound-emitting crackers between 10 pm and 6 am during Diwali.
- » **Regulations:**
 - **Noise Regulation Rules, 2000** (Amended in 2010)
 - Notified under EPA, 1986.
 - Prescribes two kind of standards: (1. Ambient Noise Level Standards 2. Noise levels for designated machinery, appliance, firecrackers etc.)
 - **Silence zone:** zones of silence (100 meters) near schools, courts, hospitals etc.
 - **Peaceful nights:** No permission for public address (PA) system in open after 10 pm and before 6 am.
- » **Monitoring:** **National Ambient Noise Monitoring Network (NANMN)** was launched in 2011

- **Way Forward:**

- » **Stricter Implementation and Enforcement:** Norms, regulations already exist, but CPCB and SPCBs lack resources and sometimes fails to prioritize that challenge of noise pollution. It's important to create deterrence by ensuring penalization.
- » **Proper Urban Planning:**
 - Separate Residential colonies from industrial area.
 - Plant green cover as buffer to reduce noise.
- » **Public Awareness:** People's participation in the form of avoiding unnecessary honking, not using very loud firecrackers etc can go a long way in fighting noise pollution.
- » **Research & Development:**
 - For quieter machineries in industries and quieter household equipment.
 - Development of affordable and soundproof wall and window technology
 - Smart Traffic management to reduce pollution due to traffic.
- » **Special Focus on Public Announcement (PA) systems:** Regulating them in religious institutions, rallies and other such events.
- » **More Resources with Urban Local Bodies and Pollution Boards:** Proper implementation, monitoring and awareness generation will not be feasible without the above measures.

- **Conclusion:** By adopting a holistic approach which includes stricter implementation of norms, proper urban planning, people's participation, R&D on new noise reduction technologies etc., India can mitigate the problems caused by noise pollution and create a quieter and healthier environment.

8. LIGHT POLLUTION

- **Example Question:**
 - » "Electric lights have revolutionized our lives, but as illumination increases, the toll on wildlife and human health is becoming harder to ignore" Elaborate [10 marks, 150 words]
- **Introduction:** Light pollution, also known as photo-pollution or luminous pollution, is the **excessive, misdirected or invasive use of artificial outdoor lighting**.
 - » For instance, **Mumbai study by environment group Awaaz Foundation** found that light levels at many parts in the cities was several times higher than the 50-60 lux. (for reading we need 20 lux. 50-60 lux is harmful for health)
- **Negative Effect of Light Pollution:**
 - i. **Human Health and Body:**
 - Disturbs circadian rhythm in humans and induces sleep disorder and other health risks like obesity, depression and diabetes.
 - Excessive light can damage retina.
 - ii. **Biodiversity:** Natural light cycle of other faunas and flora is also negatively affected. It disorients nocturnal animal and hampers their foraging and reproductive patterns.
 - Some insects are drawn towards these light which may have negative impact on food chain
 - iii. **Wastage of Energy:** This leads to overuse of fuel, which is eventually causing climate change.
 - iv. **Astronomy** is also negatively affected. Mismanaged light alters the color and contrast of the nighttime sky and eclipses starlight. It also hinders study of the universe and makes stargazing difficult.
- **How is light pollution regulated in India?**
 - » The regulation of light pollution is mostly absent in India. There are some efforts which help in reducing light pollution like Energy Conservation Building Code and Model Building By Laws which are primarily focused on energy efficiency. However, even their implementation is poor.
- **Way Forward**
 - » **Enact Laws and Regulation** dealing directly with light pollution.
 - » **City Planning** should avoid unnecessary lighting and specially avoid light focused towards sky.
 - » **Smart Light solutions** to modify street and outdoor lights, which adjust brightness and direction based on real time needs.
 - » **Awareness:** People's participation through shading of light sources and switching of lights when not needed.
 - » **Increased focus** on dealing with space debris as they hamper space observations and create other issues problems.
- **Conclusion:** By adopting holistic approach which includes enactment of laws and regulations, their proper implementation, components in city planning, smart lighting solutions and people's participation India can mitigate the problems caused by light pollution.

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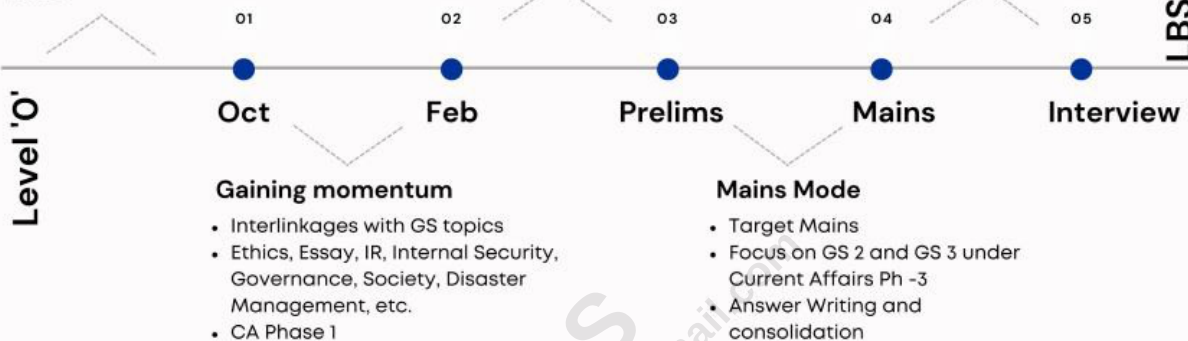
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